RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 10/11/2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Enterprise Information Systems		**Course Code : IM355TBB**

Note:
1. Answer all the questions

Sl. No	Questions	M	BT	CO
	Part – A			
1.	The evolution of planning systems progressed from MRP-I to _____ and then to ERP.	1	1	CO1
2.	The primary mechanism by which Enterprise Information Technology provides value to organizations is through appropriate _____ improvement.	1	1	CO1
3.	Enterprise Systems are noted for providing a tightly integrated solution to an organization's information needs. State two functions that were historically stand-alone applications.	2	2	CO2
4.	What is the Role of EIS in business decision-making? State one key function.	2	1	CO1
5.	A textile company is implementing a new EIS. Their goal is to maximize the utilization of their collective employee knowledge for project teaming and innovation. Which core EIS component is primarily responsible for managing and exploiting this intellectual capital? Justify your choice.	2	3	CO3
6.	What is the Role of EIS in business decision-making? State one key function.	2	1	CO1
	Part – B			
1	Distinguish between MRP-I and MRP-II. Explain the primary functions of each planning system and illustrate how the evolution from Material Requirements Planning (MRP-I) to Manufacturing Resource Planning (MRP-II) broadened the scope of enterprise planning.	10	2	CO2
2	Define and Differentiate the Core Components of EIS. Briefly define ERP, CRM, and SCM, and explain the unique focus of each component within the overall enterprise system framework.	10	2	CO2
3	Explain the scope of Digital Transformation in the Context of EIS. Describe what the term means for a modern enterprise and how the adoption of an integrated EIS serves as the foundational enabler for achieving digital transformation goals.	10	2	CO1
4	You are advising a high-growth tech startup. The founders are considering which EIS component to prioritize first: CRM for managing investor relations and early customer feedback, or a modular ERP for integrating the finance and HR functions. Argue for the strategic component that should be prioritized to support the startup's growth and eventual Digital Transformation.	10	3	CO3

5	What is a Business Process? Provide a comprehensive definition, and explain why identifying and defining these processes is the crucial first step before any automation or reengineering effort.	10	2	CO2

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**27th November 2024**	**Maximum Marks**	**10 + 50**
Course Code	**IM255TBB**	**Duration**	**120 Mins**
Sem	**V**	**CIE – I**	
ENTERPRISE INFORMATION SYSTEMS (Elective)			

Note:
1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	List two key features of a Customer Relationship Management (CRM) system.	02	L2	CO1
2	Mention two types of data commonly handled by EIS.	02	L2	CO1
3	Name two popular ERP software solutions.	02	L1	CO2
4	What is the primary function of Supply Chain Management (SCM) within EIS?	02	L1	CO1
5	What are the core modules of an ERP system?	02	L2	CO2

PART – B

Sl. No	Questions	M	BT	CO
1.	Analyse the benefits and challenges of implementing an EIS in a manufacturing organization. Support your answer with examples.	10	L2	CO1
2.	How can information technology support a company's business processes and decision making and give it competitive advantage? Give examples to illustrate your answer.	10	L3	CO3
3.	Highlight some of the issues you have faced in using SAP implemented in your institution. How you resolved those issues.	10	L3	CO2
4.	Evaluate the role of EIS in Customer Relationship Management (CRM) with specific examples of how companies have leveraged EIS for CRM.	10	L2	CO3
5.	XYZ Manufacturing Company has been facing inefficiencies in its operations due to a lack of integration between its departments. The procurement, inventory, production, and sales teams operate on standalone systems, leading to delays, data inaccuracies, and increased costs. To address these issues, the company has decided to implement an Enterprise Resource Planning (ERP) system. However, after six months of implementation, the company is still struggling with system adoption and is not seeing the expected improvements. Analyze the challenges XYZ Manufacturing is facing in implementing the ERP system. Suggest strategies to overcome these challenges and ensure successful adoption of the ERP system.	10	L4	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

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Academic year 2024-2025 (ODD Sem)

	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	06	04	–	–	–	04	06	–	–	–	–
		Test	10	10	20	10	–	–	20	20	10	–	–

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 07/11/2025	**CIE – 1**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: ADVANCED DECISION MODELLING		**Course Code : IM355TBD**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is balking, reneging and jockeying?	02	L2	CO1
2.	The Probability that exactly zero units in the queuing system is _____	02	L2	CO1
3.	Write the Kendall's notation for the single server queuing system	02	L2	CO1
4.	Classify the states of Markov Chain.	02	L2	CO2
5.	In an M/M/1 queue, the arrival is Poisson with 4 per minute and the average time spent in system is 5.5 minutes for each customer. The service rate is _____ per minute	02	L3	CO1
	Part – B			
1	Customers arrive at a one-window drive according to a Poisson distribution with mean of 10 minutes and service time per customer is exponential with mean of 6 minutes. The space in front of the window can accommodate only three vehicles including the serviced one. Other vehicles have to wait outside this space. Calculate: i. Probability that an arriving customer can drive directly to the space in front of the window. ii. Probability that an arriving customer will have to wait outside the directed space. iii. How long an arriving customer is expected to wait before getting the service?	10	L3	CO1
2	At a stamp vending window of a post office 20 customer arrive on an average every 10 min. the vendor serves 5 customers in 2 min. determine: 1. Average number of customers in the system 2. Average waiting time of a customer. 3. Probability of a customer waiting for more than 3 min before being served. 4. Idle time of the vendor in a shift of 8 hours	14	L3	CO1
3	Discuss the classification of queuing models and explain how Kendall's notation helps in representing them.	06	L3	CO1

4	In a bank there is only one window. A solitary employer performs all the services required and the window remain continuously open from 7am to 1 pm. It was discovered that an average number of clients is 54 during the day and the average service time is 5min / person. Find: i. Average number of clients in the system ii. Average waiting time iii. The probability that a client has to spend more than 10 mins in the system	10	L3	CO1
5	A system has three possible states, 0, 1 and 2. Every hour it makes a transition to a different state, which is determined by a coin flip. For example, from state 0, it makes a transition to state 1 or state 2 with probabilities 0.5 and 0.5. a) Find the transition probability matrix. b) Find the three-step transition probability matrix. c) Find the steady-state distribution of the Markov chain.	10	L3	CO2

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**27th November 2024**	**Maximum Marks**	**50**
Course Code	**IM355TBD**	**Duration**	**90 Min**
Sem	**VI**	**CIE – 1**	
ADVANCED DECISION MODELLING (Professional Core Elective)			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part A			
1.	What is the primary objective of sequencing problems?	02	1	1
2.	How does the concept of 'setup time' impact sequencing problems?	02	3	3
3.	What is the significance of 'priority rules' in sequencing problems?	02	2	2
4.	What is the difference between 'flow shop' and 'job shop' sequencing?	02	2	2
5.	How does the concept of 'setup time' impact sequencing problems?	02	3	3
	Part B			
1	Explain the following terminologies: i. Idle time on machine ii. Total elapsed time iii. No passing rule iv. Number of machines v. Processing order	10	1	1
2	Find the sequence that minimizes the total elapsed time (in hours) required to complete the following tasks on two machines. *Task* | *A* | *B* | *C* | *D* | *E* | *F* | *G* | *H* | *I* *Machine I* | 2 | 5 | 4 | 9 | 6 | 8 | 7 | 5 | 4 *Machine II* | 6 | 8 | 7 | 4 | 3 | 9 | 3 | 8 | 11	10	2	2
3	Determine the optimal sequence of jobs that minimize the total elapsed time, based on the following information. Processing time on machines is given in hours and passing is not allowed. *Job No* | *A* | *B* | *C* | *D* | *E* | *F* | *G* *Machine* M_1 | 3 | 8 | 7 | 4 | 9 | 8 | 7 *Machine* M_2 | 4 | 3 | 2 | 5 | 1 | 4 | 3 *Machine* M_3 | 6 | 7 | 5 | 11 | 5 | 6 | 12	10	3	3

Academic year 2024-2025 (ODD Sem)

<table>
<tr><td>4</td><td>There are four jobs each of which has to be processed on machines A, B, C, D, E and F in the order ABCDEF. Processing time in hours is given in the table below. Find the optimal sequencing of jobs, minimum time required to process these jobs and the idle time for each of the machines.
<table>
<tr><td>Job</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td></tr>
<tr><td>1</td><td>15</td><td>8</td><td>6</td><td>14</td><td>6</td><td>36</td></tr>
<tr><td>2</td><td>17</td><td>7</td><td>9</td><td>10</td><td>15</td><td>22</td></tr>
<tr><td>3</td><td>21</td><td>7</td><td>12</td><td>9</td><td>11</td><td>19</td></tr>
<tr><td>4</td><td>18</td><td>6</td><td>11</td><td>12</td><td>14</td><td>17</td></tr>
</table></td><td>10</td><td>2</td><td>2</td></tr>
<tr><td>5</td><td>Two major parts P_1 and P_2 for a product require processing through machine centres. The technological sequence of the parts on six machines and manufacturing times on each machine are
<table>
<tr><td rowspan="2">Part P_1</td><td>Machine sequence:</td><td>C</td><td>A</td><td>E</td><td>F</td><td>D</td><td>B</td></tr>
<tr><td>Time (hrs):</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>1</td></tr>
<tr><td rowspan="2">Part P_2</td><td>Machine sequence:</td><td>B</td><td>A</td><td>E</td><td>F</td><td>C</td><td>D</td></tr>
<tr><td>Time (hrs):</td><td>3</td><td>2</td><td>5</td><td>3</td><td>2</td><td>3</td></tr>
</table>
What would be the optimal scheduling to minimize the total processing time for these two parts? Find also the total elapsed time. For each machine specify the part that should be processed first.</td><td>10</td><td>3</td><td>3</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	1	2	2	–	1	2	2	–	–	–
	Test	Max Marks	10	20	20	–	10	20	20	–	–	–

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